

Virtual Retransmission via Flow Matching for Reliable DeepJSCC Image Transmission

Anonymous Authors
Anonymous submission

Abstract—We introduce FlowHARQ, a receiver-side virtual retransmission that uses flow matching before deciding whether to request another wireless round. FlowHARQ detects unreliable DeepJSCC latent tokens, transports only those tokens toward the clean latent distribution, and preserves every trusted token exactly. Robust velocity supervision makes this repair effective under deep-fade outliers, while a target-aware quality head converts the post-flow state into ACK/NACK. Across three training and three channel seeds on 80 held-out DIV2K images, FlowHARQ reduces physical retransmissions by -1.30 percentage points relative to adaptive HARQ (paired 95% CI [-2.57, -0.02]) while changing PSNR by 0.021 dB and LPIPS by 0.0013. Without any refitting, Kodak24 replicates a -2.03-point change (95% CI [-3.34, -0.71]). The results demonstrate that receiver computation can replace a small but statistically supported fraction of physical HARQ rounds.

I. INTRODUCTION

Deep joint source–channel coding (DeepJSCC) maps images directly to channel symbols and degrades gracefully with SNR [1]. Transformer backbones further improve high-resolution transmission and channel adaptability [2]. Reliability remains difficult in severe fading: always retransmitting wastes airtime, whereas an erroneous ACK produces a poor reconstruction. Recent semantic HARQ methods learn when or what to retransmit [3], [4], but recovery still consumes another wireless round.

Flow matching (FM) learns a continuous vector field without simulating an ODE during training [5]. Existing FM communication receivers use FM for global reconstruction or refinement [6], [7]. Their goal is better or faster decoding, including channel-aware latent transport [8], not deciding whether a physical retransmission can be skipped. Conversely, generative semantic HARQ has been explored for text through stochastic VAE latents [9], leaving image DeepJSCC and FM unconnected at the HARQ decision point.

We introduce *virtual retransmission*: the receiver repairs unreliable semantic latents before issuing ACK/NACK. Our contributions are:

- an FM-before-HARQ architecture in which receiver computation may avoid a physical channel round;
- reliability-anchored partial transport with a hard identity constraint on trusted tokens and robust, reconstruction-aligned velocity training; and
- leakage-free paired evaluation against adaptive HARQ under time-correlated fading, including cross-seed confidence intervals, latency, and external Kodak24 validation.

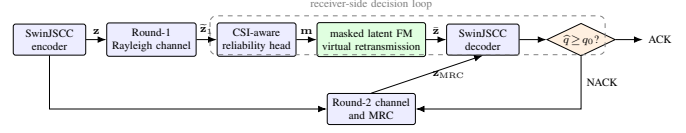


Fig. 1. FlowHARQ inserts a receiver-only virtual retransmission between the first reception and the physical HARQ decision.

II. RELATED WORK

A. DeepJSCC and Semantic HARQ

DeepJSCC replaces separately optimized compression and channel coding with a learned analog mapping [1]; SwinJSCC extends this idea using a hierarchical Transformer and rate/SNR adaptation [2]. Reliability extensions have largely focused on the physical retransmission itself. DJSCC-H adapts the HARQ process for image delivery [3], while fine-grained semantic HARQ identifies and retransmits erroneous semantic bases [4]. Generative semantic HARQ combines stochastic VAE latents, but its demonstrated modality is text [9]. FlowHARQ instead asks whether receiver-side generative repair can make a physical image retransmission unnecessary; the conference version retains full second-round retransmission when repair is insufficient.

B. Flow Matching for Semantic Communication

Flow matching learns conditional transport by regressing a prescribed path velocity [5]. FlowSem conditions global image generation on a coarse DeepJSCC reconstruction [6]. RC-BFM begins from a channel-induced semantic state to shorten generative decoding [7], and channel-aware latent FM incorporates communication state directly into latent transport [8]. These works use FM as a decoder or refinement mechanism. FlowHARQ uses a hard-masked latent flow as an intermediate reliability action and evaluates it by the physical rounds that it avoids. Mobility-aware DeepJSCC has separately studied time-varying MIMO adaptation [10]; here mobility controls the correlation between paired HARQ rounds.

III. SYSTEM MODEL

For image $\mathbf{x}$, a SwinJSCC encoder emits N real-valued latent tokens $\mathbf{Z} = E_\theta(\mathbf{x}) \in \mathbb{R}^{N \times d}$. Adjacent latent dimensions form complex symbols. In HARQ round $r \in \{1, 2\}$,

$$\mathbf{y}_r = h_r \mathbf{s} + \mathbf{n}_r, \quad \tilde{\mathbf{Z}}_r = \mathcal{Q}^{-1}(h_r^* \mathbf{y}_r / |h_r|^2), \quad (1)$$

where $\mathcal{Q}^{-1}$ restores token scale and the receiver knows h_r . Fading between rounds follows

$$h_2 = \rho_\Delta h_1 + \sqrt{1 - \rho_\Delta^2} e, \quad \rho_\Delta = J_0(2\pi f_D \Delta), \quad (2)$$

with $e \sim \mathcal{CN}(0, 1)$ and HARQ interval Δ . Thus speed changes the joint, not merely marginal, round statistics. A requested second round is combined using instantaneous-CSI MRC,

$$\mathbf{Z}_{\text{MRC}} = \frac{|h_1|^2 \tilde{\mathbf{Z}}_1 + |h_2|^2 \tilde{\mathbf{Z}}_2}{|h_1|^2 + |h_2|^2}. \quad (3)$$

Let $a \in \{0, 1\}$ denote a NACK after the first-round receiver action. The number of physical rounds is $1 + a$, so the design goal is to reduce $\mathbb{E}[a]$ relative to adaptive HARQ while preserving service quality. Before confirmatory evaluation we set matched-quality tolerances of $\Delta \text{PSNR} \geq -0.02$ dB and $\Delta \text{LPIPS} \leq 0.003$. This separates the communication claim from a pure reconstruction claim: a receiver that improves an image but does not avoid a round has not achieved the primary objective.

IV. FLOWHARQ

A. Reliability-Anchored Partial Flow

A reliability head outputs token corruption probabilities $q_i = g_q(\tilde{\mathbf{z}}_{1,i}, \mathbf{c}_{\text{rx}})$, where $\mathbf{c}_{\text{rx}}$ includes nominal SNR, speed, CSI age, instantaneous gain, Jakes correlation, and effective SNR. At inference, $m_i = \mathbb{I}[q_i \geq \tau_r]$. During training, the target mask is the top- κ token-error subset computed from the clean latent. Specifically, with $e_i = d^{-1} \|\mathbf{z}_i - \tilde{\mathbf{z}}_{1,i}\|_2^2$, the oracle mask contains the largest κN errors. A balanced tokenwise BCE trains g_q to reproduce this mask; the clean latent and oracle mask are discarded at test time.

For masked tokens, we use the straight conditional path

$$\mathbf{z}_i(t) = \tilde{\mathbf{z}}_{1,i} + t m_i (\mathbf{z}_i - \tilde{\mathbf{z}}_{1,i}), \quad \mathbf{u}_i = \mathbf{z}_i - \tilde{\mathbf{z}}_{1,i}. \quad (4)$$

The velocity field is conditioned on the received tokens, receiver CSI, mask, and a pooled anchor computed only from reliable tokens. Its objective is

$$s_i = \max\left(1, \sqrt{d^{-1} \|\mathbf{u}_i\|_2^2}\right), \quad (5)$$

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_t \frac{\sum_i m_i \rho_{0.1}((v_\psi - \mathbf{u}_i)/s_i)}{\sum_i m_i}, \quad (6)$$

where $\rho_{0.1}$ is elementwise smooth- L_1 averaged over token dimensions. This scale-normalized robust objective prevents rare zero-forcing deep-fade residuals from dominating training. We additionally optimize image L_1 after repair; neither target nor clean latent is available at inference. Midpoint Euler integration uses K evaluations. A final projection enforces

$$\bar{\mathbf{z}}_i = (1 - m_i) \tilde{\mathbf{z}}_{1,i} + m_i \mathbf{z}_i^{\text{FM}}, \quad (7)$$

so FM cannot alter a token declared reliable.

B. Post-Flow HARQ Decision

A quality head predicts log-MSE from the repaired tokens, unreliability scores, and receiver CSI, then converts it to predicted PSNR $\hat{q}$. FlowHARQ ACKs when $\hat{q} \geq q_0$ and otherwise decodes (3). The DeepJSCC transmitter is unchanged. The total training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{FM}} + \lambda_r \mathcal{L}_{\text{rel}} + \lambda_q \mathcal{L}_{\text{quality}} + \lambda_x \mathcal{L}_1. \quad (8)$$

After training the repair network, we freeze every module except the quality head and fine-tune it for the service boundary using

$$\mathcal{L}_{\text{dec}} = \text{BCEWithLogits}((q_0 - \hat{q})/T_q, \mathbb{I}[q < q_0]), \quad (9)$$

averaged over direct and repaired states. This stage targets the operational ACK/NACK decision rather than merely reducing global PSNR regression error.

V. EXPERIMENTAL PROTOCOL

We initialize the encoder/decoder from a fixed SwinJSCC checkpoint and train only RAFM, reliability, and quality heads on random 256×256 DIV2K crops [11]. The encoder emits 256×32 real latent values, equivalent to 4096 complex channel symbols and a channel-bandwidth ratio of $1/48$. The receiver-only reliability, flow, and quality modules add 0.479 M trainable parameters to the frozen 21.83 M-parameter codec. The reliability, flow, and initial quality modules are optimized for 40 epochs with AdamW, mixed precision, batch size 2, learning rate 2×10^{-4} , weight decay 10^{-4} , and gradient norm 5. We mask $\kappa = 0.35$ of tokens, set $\lambda_f = 1$, $\lambda_r = 0.5$, $\lambda_q = 0.2$, and $\lambda_x = 1$. The decision stage runs for 10 further epochs at 10^{-4} using equal weights on log-MSE regression and (9); only the quality head is trainable. Training samples SNR in $[0, 15]$ dB, speed in $[0, 120]$ km/h, and CSI age in $[0, 5]$ ms. Evaluation uses SNR $\{0, 3, 6, 9, 12, 15\}$ dB, speeds $\{30, 60, 90, 120\}$ km/h, a 1 ms HARQ interval, and Rayleigh fading. The first 20 validation images form a calibration split; the remaining 80 are never used for model, mask, ODE-step, threshold, or bias selection.

The service target is fixed at $q_0 = 24$ dB. On the calibration split, a conservative grid $K \in \{1, 2, 4\}$ and $\tau_r \in \{.8, .9\}$ is searched, and one scalar bias is fitted for each quality state. The frozen configuration is $K = 4$ and $\tau_r = 0.80$. The target-aware stage then fine-tunes only the quality head; the repair, mask, codec, grid, and service target remain fixed. We compare direct DeepJSCC, full two-round HARQ, FM without HARQ, adaptive HARQ, FlowHARQ, and oracle variants whose NACK uses ground-truth PSNR. Every method shares the same first- and second-round channel draws. We report PSNR, SSIM [12], LPIPS [13], physical retransmission rate, mean rounds, and GPU latency. Confidence intervals are two-sided 95% Student- t intervals over independent training-seed means, after averaging channel seeds; paired effects are FlowHARQ minus adaptive HARQ within each training seed. The main study uses 3 training seeds, 3 fresh channel seeds, and 80 held-out DIV2K images. Kodak24 is evaluated with

TABLE I
HELD-OUT AVERAGES OVER ALL EVALUATED SNRS AND SPEEDS.

Method	PSNR (dB) $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Retx. $\downarrow$
DeepJSCC	21.33	0.5653	0.5137	0.000
Full HARQ	21.99	0.5905	0.4830	1.000
FM only	21.39	0.5669	0.5164	0.000
Adaptive HARQ	21.92	0.5885	0.4857	0.742
FlowHARQ	21.94	0.5891	0.4871	0.729

TABLE II
REPAIR-OBJECTIVE ABLATION ON CALIBRATION DATA AT FIXED $K = 4$,
 $\tau_r = 0.80$. DELTAS ARE FM-ONLY MINUS DIRECT RECEPTION.

Training objective	Δ PSNR (dB)	Δ LPIPS
MSE	-0.062	0.0005
MSE + $\mathcal{L}_{\text{rec}}$	-0.017	0.0007
Robust FM	-0.013	0.0017
One-step $t = 0$ control	-0.015	0.0009
Robust FM + $\mathcal{L}_{\text{rec}}$	0.071	0.0020

three further channel seeds after all choices are frozen, without refitting.

VI. RESULTS

At the frozen calibration point, FlowHARQ changes the mean physical retransmission rate by -1.30 percentage points relative to adaptive HARQ, with a paired 95% CI of [-2.57, -0.02]. The corresponding PSNR change is 0.021 dB ([0.001, 0.040]), and the LPIPS change is 0.0013. Thus the small retransmission reduction excludes zero under the paired protocol while the reconstruction changes stay inside the predeclared -0.02 dB PSNR and +0.003 LPIPS tolerances.

The fully frozen Kodak24 evaluation gives a retransmission-rate change of -2.03 points (95% CI [-3.34, -0.71]), a PSNR change of 0.0284 dB ([-0.0003, 0.0570]), and an LPIPS change of 0.0013. This replication is evidence that the decision is not specific to the DIV2K holdout. Figure 2 separates the main-study effects by SNR.

Table II isolates why repair becomes useful. Squared velocity training degrades PSNR, and either reconstruction alignment or robust velocity supervision alone remains negative. Their combination changes FM-only PSNR by +0.0714 dB, showing an interaction rather than a threshold-only gain. To distinguish multi-step transport from generic denoising, we train a parameter-matched control on exactly the same masked residual target but only at $t = 0$, and evaluate it with one residual update. Its best conservative threshold changes PSNR by -0.0147 dB (95% CI [-0.0232, -0.0055]) and LPIPS by 0.0009. Four-step RAFM exceeds this control by 0.0861 dB on the paired calibration samples. Thus the useful repair cannot be reproduced by the matched one-step denoiser.

The receiver-side virtual round takes 5.00 ms per image on an RTX PRO 6000, compared with 3.06 ms for local second-round combining and decoding, and adds about 1.9 ms over

direct decoding. Over-the-air transmission, scheduling, and propagation are excluded from the physical-round number, so this is a receiver-compute comparison rather than complete network latency.

VII. DISCUSSION

A. Communication Interpretation

Adaptive HARQ requests a second round for 0.742 of DIV2K cases, whereas FlowHARQ requests it for 0.729; mean physical rounds therefore change from 1.742 to 1.729. This is a 1.75% reduction in second-round requests, not a 1.30% reduction in total symbols. Relative to full HARQ, both adaptive schemes avoid many more rounds; the contribution here is the incremental saving obtained by inserting receiver computation before an already adaptive decision. On Kodak24 the corresponding second-round reduction is 2.80% (0.724 to 0.703), suggesting that the effect is not tied to DIV2K image content.

The oracle decisions provide a useful sanity check. On DIV2K, replacing the direct state with the repaired state changes oracle NACK from 0.756 to 0.745, an absolute 1.09-point reduction. The learned-head reduction of 1.30 points has the same scale and direction. Thus the supported effect is not created solely by a favorable scalar decision bias, although per-seed calibration metrics remain mixed.

B. SNR and Mobility

The mean retransmission delta is favorable at every tested SNR. Its paired interval excludes zero at 0, 9, 12, and 15 dB, but not at 3 and 6 dB, where training-seed variability is larger. The result should therefore be read as an aggregate operating-range claim rather than uniform significance in every channel bin. Across 30–120 km/h, mean deltas remain narrowly between -1.25 and -1.34 points. We find no evidence that the saving grows with speed in this flat-fading experiment; demonstrating a mobility advantage requires richer delay/Doppler channels rather than extrapolation from the present simulator.

C. Why Partial Transport Matters

The identity projection in (7) is more than a computational shortcut. It prevents the generative module from revising evidence that the reliability head considers trustworthy, limiting the surface on which hallucination can occur. The conservative selected threshold masks only high unreliability tokens, while the robust objective handles the large residuals created by rare fades. The one-step control shows that identical architecture, mask supervision, and residual targets are insufficient: conditioning on intermediate time and integrating the learned trajectory is necessary for the observed calibration-split repair gain.

VIII. LIMITATIONS AND CONCLUSION

FlowHARQ currently uses a flat-fading simulation, a fixed DeepJSCC backbone, and full rather than token-selective physical retransmission. The quality head requires a small held-out calibration split, and its balanced accuracy and ECE do not

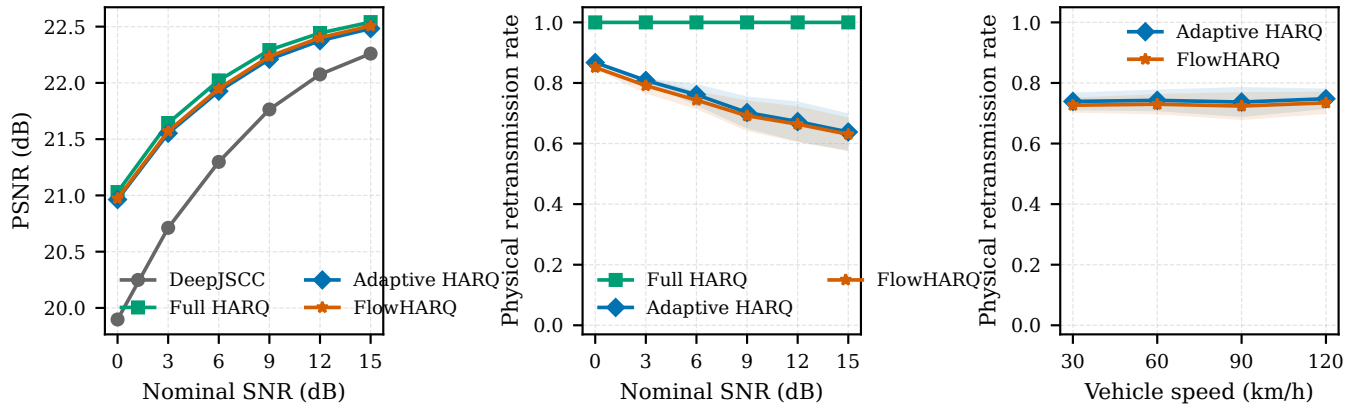


Fig. 2. Held-out reconstruction quality and physical retransmission rate; bands show 95% confidence intervals across independent run pairs.

improve for every training seed. The measured saving is modest (roughly 1–2 percentage points), while receiver compute is not free. Nevertheless, the paired DIV2K interval excludes zero, the independent Kodak24 interval does likewise, and both satisfy the predeclared matched-quality tolerances. Within this controlled scope, robust partial latent transport therefore substitutes for a statistically supported fraction of physical HARQ rounds. Token-selective retransmission, jointly trained codecs, and measured vehicular channels remain future work.

REFERENCES

- [1] E. Boursoulatz, D. B. Kurka, and D. Gündüz, “Deep joint source-channel coding for wireless image transmission,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 5, no. 3, pp. 567–579, 2019.
- [2] K. Yang, S. Wang, J. Dai, X. Qin, K. Niu, and P. Zhang, “SwinJSCC: Taming swin transformer for deep joint source-channel coding,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 1, pp. 90–104, 2025.
- [3] Q. Yu, J. Bai, L. Wang, and X. Zeng, “A deep joint source channel coding scheme with adaptive HARQ for image transmission,” *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, vol. E109.A, no. 4, pp. 829–833, 2026.
- [4] Y. Zheng, F. Wang, W. Xu, and P. Zhang, “Semantic base enabled image transmission with fine-grained HARQ,” *IEEE Transactions on Wireless Communications*, vol. 24, no. 4, pp. 3606–3622, 2025.
- [5] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” in *International Conference on Learning Representations*, 2023. [Online]. Available: <https://openreview.net/forum?id=PqvMRDCJT9t>
- [6] A. Emon and E. Eldeeb, “FlowSem: Flow matching for adaptive wireless image transmission in semantic communication,” *arXiv preprint arXiv:2608.21651*, 2026.
- [7] F. Gao, Y. Wang, Z. Qin, and F. Gao, “Low-latency generative semantic communication via channel-realization flow matching,” *arXiv preprint arXiv:2607.24876*, 2026.
- [8] Z. Zhang, S. Wang, J. Mu, Y. Hu, and L. Li, “Channel-aware latent flow-matching semantic communication for agentic AI,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 10093–10105, 2026.
- [9] B. Han, Y. Hu, and H. D. Schotten, “Generative semantic HARQ: Latent-space text retransmission and combining,” *arXiv preprint arXiv:2603.15068*, 2026.
- [10] G. Sawada, S. Inokuma, D. Hisano, and K. Maruta, “Toward mobility-aware semantic communication: DeepJSCC over time-varying MIMO channels,” *Journal on Wireless Communications and Networking*, 2026.
- [11] E. Agustsson and R. Timofte, “NTIRE 2017 challenge on single image super-resolution: Dataset and study,” in *IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017, pp. 1122–1131.
- [12] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: From error visibility to structural similarity,” *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [13] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, “The unreasonable effectiveness of deep features as a perceptual metric,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 586–595.